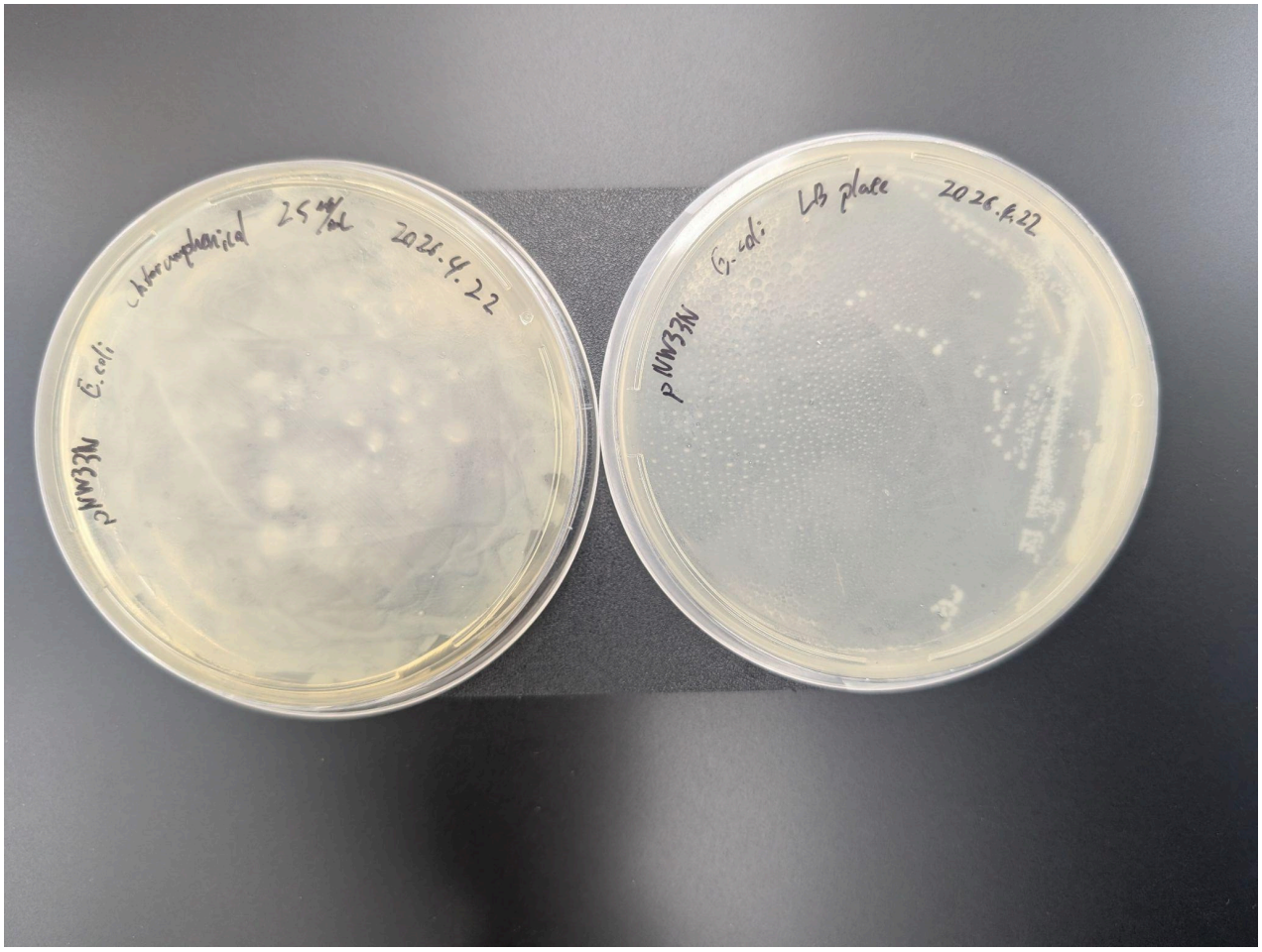
Date: 20260422
Title: Recover E. coli harboring pNW33N from BCRC
Purpose: Recover *E. coli* harboring the shuttle vector pNW33N, which was obtained from BCRC.
Procedure:

1. Heat the top of the glass vial with the Bunsen burner.
2. Add sterile water and crack it with tweezers.
3. Remove paper and cotton.
4. Add 500 μL LB medium and resuspend the tablet.
5. Drop the supernatant and streak it with four-quadrant streak method.
6. Spread 100 μL supernatant on LB plate with 100 μL 25 mg/mL chloramphenicol.
7. Inoculate the remaining supernatant in 5 mL LB medium with 25 μg/mL chloramphenicol.
8. Incubate in 37℃, started from 18:00

Done by Chars, recorded by Chars.

Results:



Recorded by Chars

Date: 20260423
Title: Miniprep
Purpose: Using miniprep kit (Viogene) to isolate pNW33N plasmid from *E. coli*
Procedure:

1. Collected cells from liquid culture by centrifuge 23444 g for 1 min.
2. Repeat the previous step until all of the liquid cultures were collected.
3. Add 200 μL MX1 buffer and resuspend pellet by pipetting.
4. Add 250 μL MX2 buffer and gently invert 8 times.
5. Incubated at room temperature for 2 min.
6. Add 350 μL MX3 buffer and gently invert 8 times.
7. Centrifuge at 23444 g for 5 min.
8. Transfer the supernatant to the column by P200 micropipette.
9. Centrifuge at 23444 g for 30 sec.
10. Discard the flow-through. Add 500 μL WN buffer.
11. Centrifuge at 23444 g for 30 sec.
12. Discard the flow-through. Add 700 μL WS buffer.
13. Centrifuge at 23444 g for 30 sec.
14. Discard the flow-through. Centrifuge the empty column at 23444 g for 3 min.
15. Add 50 μL EB buffer at 65℃. Incubated at room temperature for 5 min.
16. Centrifuge at 23444 g for 3 min.
17. Measure the concentration by Biodrop

Results:

Plasmid	Concentration (ng/μL)	A260/280	A260/230
pNW33N #1	71	1.775	0.910
pNW33N #2	78	1.660	0.684

Done by Chars, recorded by Chars.

Date: 20260423
Title: Enzyme Digestion
Purpose:

(1) Verify pNW33N obtained from BCRC
(2) Prepare vector for Gibson assembly

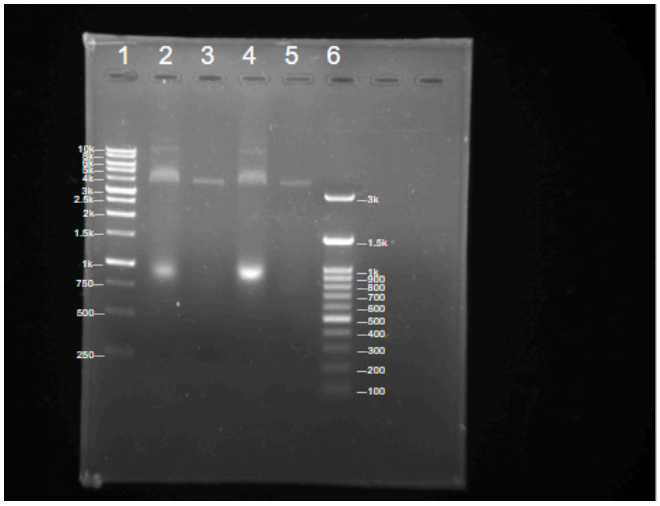
Procedure:

1. Prepare the reaction.

	#1 (μL)	#2 (μL)
Vector (1 μg)	14.1	12.8
10x CutSmart buffer	5	5
Distilled water	28.9	30.2
EcoRI-HF	1	1
BtsI-v2	1	1

2. Incubate in a PCR machine at 37℃ for 2 hours.
3. Resolve in 1% agarose gel in 1x TAE for 30 min.

Results:



Lane	Sample	Volume(μL)	DNA Amount(ng)
1	Omics 1 kb DNA RTU Ladder	5	
2	pNW33N #1	5	355

3	pNW33N #1, EcoRI/BtsI digestion, 2 hr	5	100
4	pNW33N #2	5	390
5	pNW33N #2, EcoRI/BtsI digestion, 2 hr	5	100
6	[DM2300] ExcelBand™ 100 bp+3K DNA Ladder	5	

Done by Chars, recorded by Chars

Date: 20260427
Title: Hydroponic System Testing
Purpose:Prepare hydroponic system for testing
Preparation of 40 ml ½ MS agar
Procedure:

1. 1/2MS agar preparation
    a. 1.58 g MS powder _ 40 mL ddH2O
    b. Sterilize the mixture using autoclave
2. Prepare the holder (1.5ml tube lid)
    a. Use a leather punch to make 3-mm holes in the lid.
    b. Pour the agar into 1.5 ml eppendorf's lids and wait for solidification.
3. Sow one seed in each hole of the lid. Vernalization
    a. Put the seed-containing holder at 4°C for 3 days (wrap with foil)

Done by Gabriel, Recorded by Sophie L

Date: 20260427
Title: Preparation of PCR-Amplified sfGFP fragments
Purpose:to create high-yield, stable DNA inserts for downstream molecular biology applications.
Procedure:

1. Prepare the reaction as follows:

5x Q5 Reaction buffer	10 µL
2.5 mM dNTP mix	4 µL
10 µM pNW33N-L2(sfGFP) F primer	2.5 µL
10 µM pNW33N-L2(sfGFP) R primer	2.5 µL
70 µg/mL pJUMP49-2B plasmid	1 µL
Q5 Hi-Fi DNA polymerase	0.5 µL
ddH_2O	29.5 µL
Total	50 µL

2. PCR program

Denaturation	98°C	30 sec
Denaturation (25 cycles)	98°C	10 sec
Annealing (25 cycles)	60°C	30 sec

Extension (25 cycles)	72°C	1 min
Extension	72°C	2 min

3. Storage PCR product at 4°C
4. PCR product were confirmed by gel electrophoresis (Amplicon size:1211 bp)

Done by Gabriel, Recorded by Sophie

Date: 20260428
Title: Cleanup of PCR or Restriction enzyme digestion products Purpose: Remove contaminants Mix the PCR product (or enzyme digestion product) with 0.5 mL GP buffer

1. Transfer the mixture into GP column
2. Centrifuge 1 min at 7000g, then discard the flow-through
3. Add 0.5 mL WN buffer into the column
4. Centrifuge 1 min at 7000g, then discard the flow-through
5. Add 0.5 mL WS buffer into the column
6. Centrifuge 1 min at 7000g, then discard the flow-through
7. Centrifuge the column at 23444g for 3 min
8. Place the column onto a 1.5 mL centrifuge tube
9. Add 30 µL elution buffer onto the membrane and wait for 3 min
10. Centrifuge at 23444g for 2 min
11. Storage at -20°C

Done by Gabriel, Recorded by Alex

Date: 20260428
Title: Gibson assembly and Transformation
Purpose: To create the shuttle vector with MoClo 2.0 system
Procedure:

1. Prepare the reaction

Component	20µL reaction	Length (base pair)
pNW33N enzyme digestion product(cleaning up) 3µg/mL	8.26µL	4021bp
pJCMP49-2B PCR product (cleaning up) 60µg/mL	0.24µL	1207bp
NEBuider HiFi DNA assembly master mix	10µL	
ddH_2O	1.5µL	
total	20µL	

2. Incubate the reaction in PCR machine at 50℃ for 15 min
3. Transfer 10µL of the product into 100µL competent cells.
4. On ice for 20 min
5. Heat shock at 42℃ for 60sec
6. Put the competent cell/DNA mixture on ice for 2 min
7. Add 1mL SOC medium and culture at 37℃ for 1 hr
8. Transfer 100µL culture medium on the LB agar plate with chloramphenicol
9. Spread the transformants on agar plate The plate incubated at 37℃ overnight
   Done by Gabriel

Date: 20260428
Title: Gibson assembly and transformation
Purpose: Create the shuttle vector with MoClo 2.0 system
Procedure:

1. Prepare reaction

substance	amount
pNW33N enzyme digestion product (cleaning up) (3μg/mL)	8.26 μL (4021 bp)
pJUMP49-2B PCR product (cleaning up) (60μg/mL)	0.24 μL (1207 bp)
NEBuider HiFi DNA assembly master mix	10 μL
ddH2O	1.5 μL
total	20 μL

2. Incubate the reaction in PCR machine at 50°C for 15 min
3. Transfer 10 μL of the product into 100 μL competent cells
4. Stand on ice for 20 min
5. Heat shock at 42°C for 60 sec
6. Put the reaction on ice for 2 min
7. Add 1 mL SOC medium and culture at 37°C for 1hr
8. Transfer 100 μL culture medium on the LB agar plate with chloramphenicol
9. Spread the liquid
10. The plate were incubated at 37°C overnight

Done by Gabriel, Recorded by Alex

Date:20260429
Title: Recover Bacillus Subtilis 168 from BCRC
Purpose: Recover Bacillus Subtilis 168 from BCRC
Procedure:

1. Heat the top of glass vial with bunsen burner
2. Add sterile water and crack it with tweezer
3. Remove paper and cotton
4. Add 500μL LB medium and resuspend the tablet
5. Spread the supernatant with four-quadrant streak method
6. Inoculate the remaining supernatant in LB medium 5mL
7. Incubate at 37°C, started from 21:30

Done by Chars, recorded by Alex

Date: 20260430
Title: Miniprep
Purpose: Isolate pNW33N - L2(sfGFP) - v1 C#1 & G#2 from DH5a
Procedure:

1. Collected cells from liquid culture by centrifuge 23444g 1min
2. Repeat the previous step until all liquid culture are collected
3. Add 200μL MX1 buffer and resuspend pellet by pipetting
4. Add 250μL MX2 buffer and gently invert 10 times
5. Incubate at room temperature for 2 min
6. Add 350 MX3 buffer and gently invert 10 times
7. Centrifuge 23444g for 5 min
8. Transfer supernatant to the column by P200
9. Centrifuge 30 sec at 7000g, then discard the flow-through
10. Add 0.5 mL WN buffer into the column
11. Centrifuge 30 sec at 7000g, then discard the flow-through
12. Add 0.7 mL WS buffer into the column
13. Centrifuge 30 sec at 7000g, then discard the flow-through
14. Centrifuge empty column at 10000g for 3 min
15. Add 50 μL of EB buffer preheated to 65°C. Incubate at room temperature for 5 min. Centrifuge at 10000g for 3min
16. Measure the concentration by Biodrop

Result:

Plasmid	Concentration (ng/μL)	A260/280	A260/230
pNW33N - L2(sfGFP) #1	41	2.050	1.708
pNW33N - L2(sfGFP) #2	43	1.955	1.720

Done by Chars, Recorded by Alex

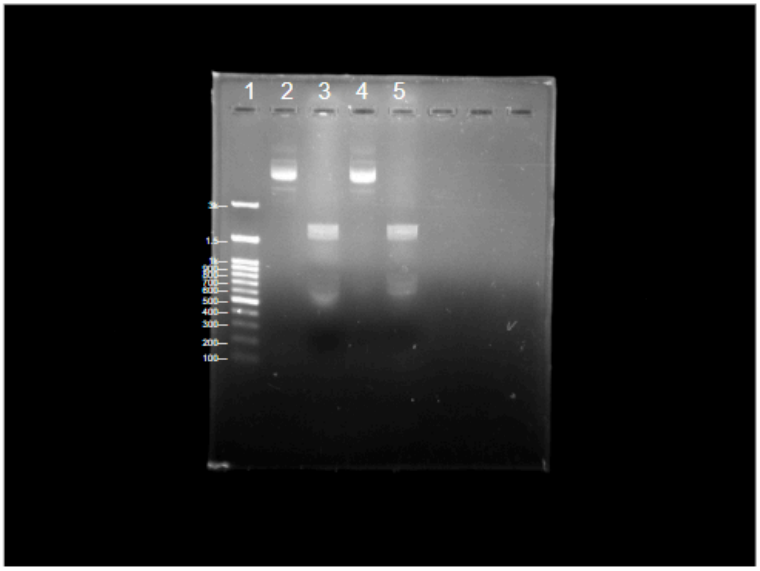
Date: 20260430
Title: Enzyme digestion
Purpose: Verify pNW33N - L2(sfGFP) from Gibson assembly
Procedure:

1. Prepare the reaction

Component	C#1	G#1
DNA (500ng)	12.2	11.6
10X CutSmart	2	2
water	5.3	5.9
BsaI - HF	0.5	0.5

2. Incubate in PCR machine at 37°C for 2hr
3. Resolve in 1% Agarose gel

Result:



Lane	Sample	Treatment	Volume(μL)	DNA Amount(ng)
1	[DM2300] ExcelBand™ 100 bp+3K DNA Ladder		5	
2	pNW33N-L2(sfG	Undigested	5	205

	FP)-v1 C#1			
3	pNW33N-L2(sfGFP)-v1 C#1	Bsal digestion, 2hr	10	250
4	pNW33N-L2(sfGFP)-v1 G#1	Undigested	5	215
5	pNW33N-L2(sfGFP)-v1 G#1	Bsal digestion,2hr	10	250

Done by Chars, recorded by Alex